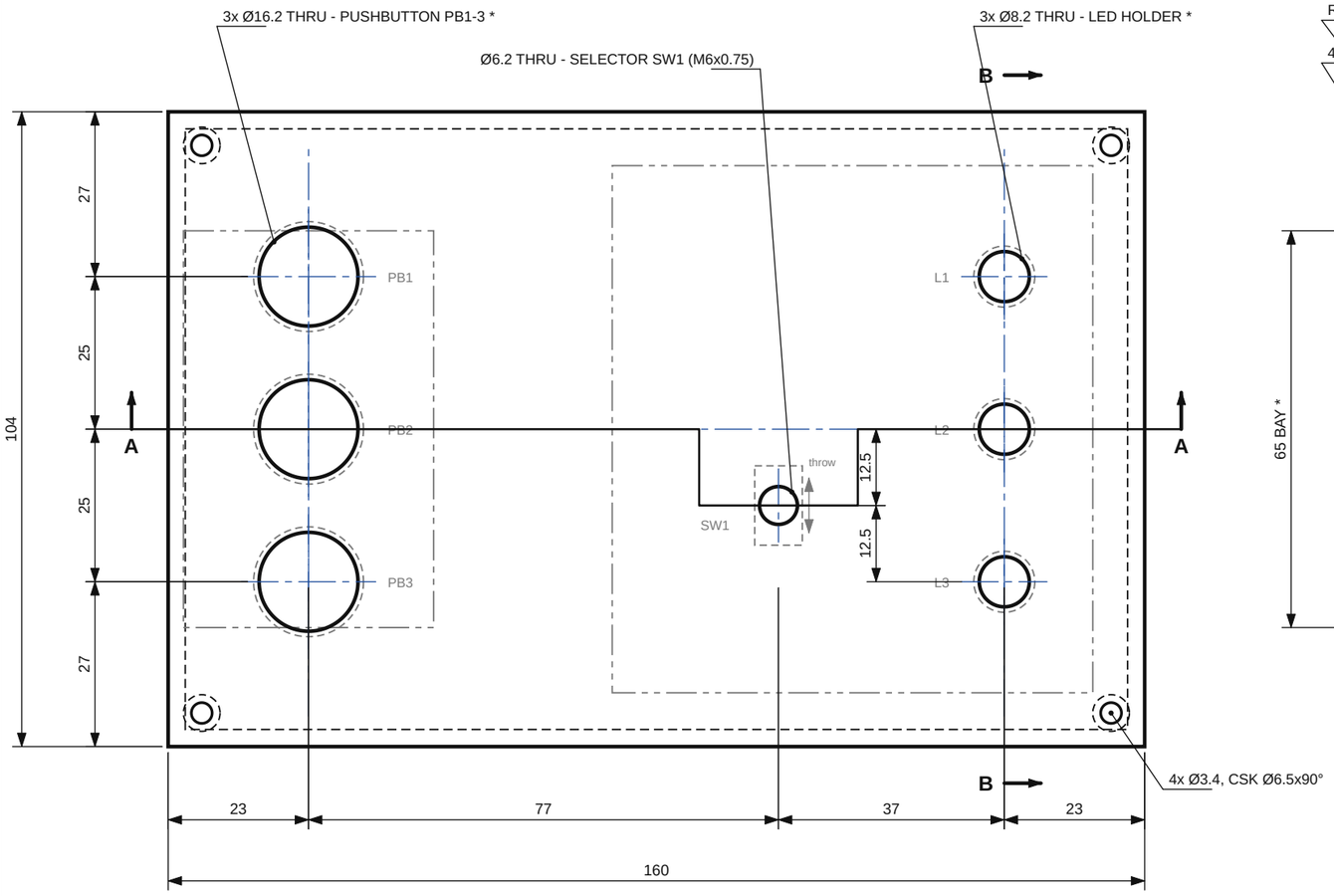
VIEW 1 - TOP (CONTROL PANEL / LID)

Lid 3.0 thick. Hidden: lid locating lip. Phantom: bays below.



VIEW 2 - PLAN, LID REMOVED

Battery bay LEFT under buttons | electronics bay RIGHT under LEDs

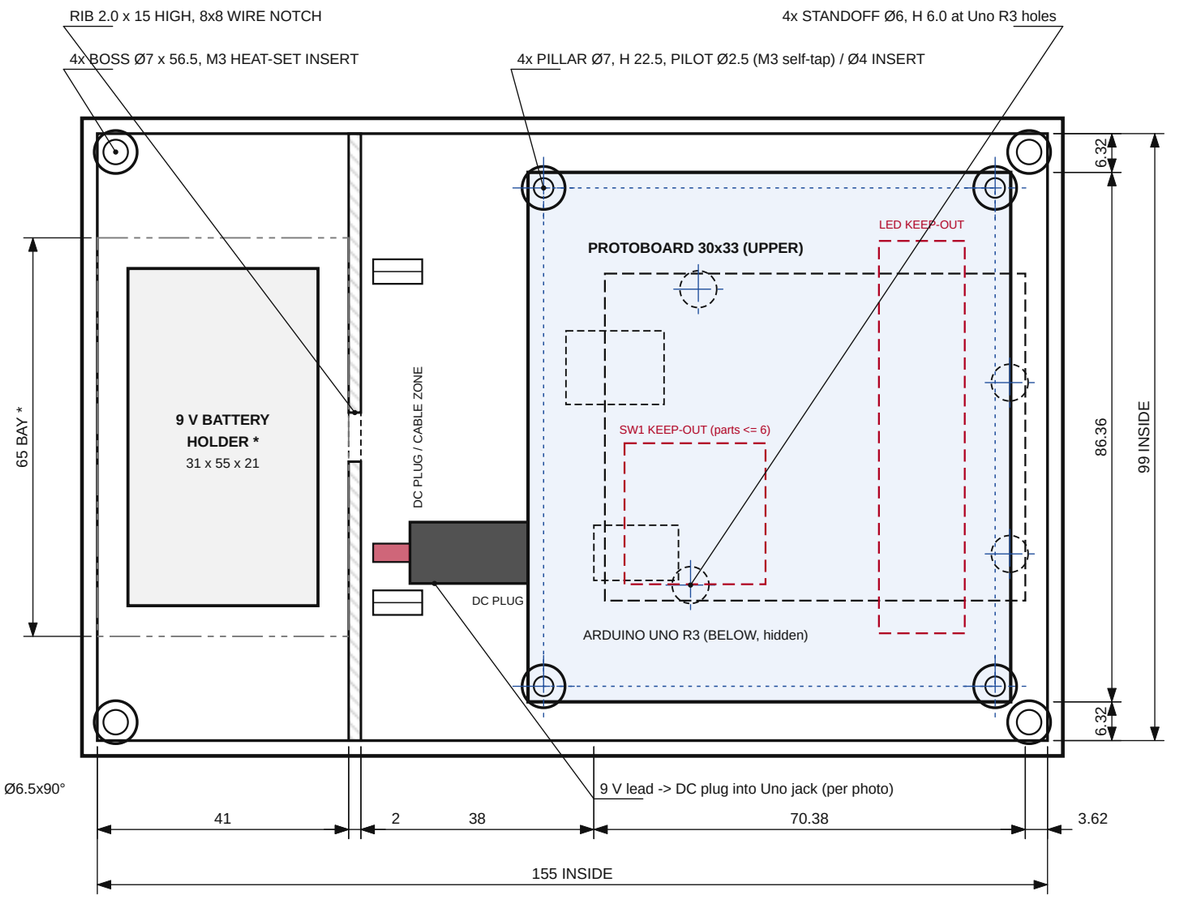


TABLE 1 - FEATURE COORDINATES (origin: external front-left-bottom corner, mm)

ITEM	X	Y	SIZE	NOTE
PB1/PB2/PB3	23	77 / 52 / 27	Ø16.2	Adafruit 1439, 16 mm illum. momentary *
SW1 selector	100	39.5	Ø6.2	Miyama MS-500E (ON)-OFF-(ON), M6x0.75
L1/L2/L3	137	77 / 52 / 27	Ø8.2	LED + 8 mm panel holder (model TBC) *
Lid screws	5.5 / 154.5	5.5 / 98.5	Ø3.4	4x M3 x 10 countersunk into inserts
Battery holder	7.5-38.5	24.5-79.5	31x55x21	on floor; VHB tape or 2x M3 *
Protoboard	72.76-151.5	8.82-95.18	78.74x86.36	underside Z 25 *
PB pillars	75.3 / 148.96	11.36 / 92.64	Ø7 x 22.5	corner holes drilled to Ø3.2
Arduino Uno R3	85.3-153.88	25.33-78.67	68.58x53.34	underside Z 8.5, USB/DC jack face LEFT
Uno standoffs	99.27 / 100.54 / 151.34 / 151.34	27.87 / 76.13 / 32.95 / 60.89	Ø6 x 6.0	4 holes, ref. Uno R3 drawing

** = nominal / assumed value - see Sheet 3, Table 4. Uno standoff X and Y are listed pairwise in the same order.*

LINE LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out / reserved

KEY DECISIONS (spec §1)

- LEFT: 9 V battery under button column
- RIGHT: protoboard above Uno R3, under LEDs
- SW1 right of centre, between L2/L3 rows (v6)
- Uno jacks face battery (photo); no USB hole
- Envelope: 160 x 104 x 62 (was 200x120x75)

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 to layout-v6 position.
C 2026-11-09 Uno + 9 V plug per photo. * = assumed.

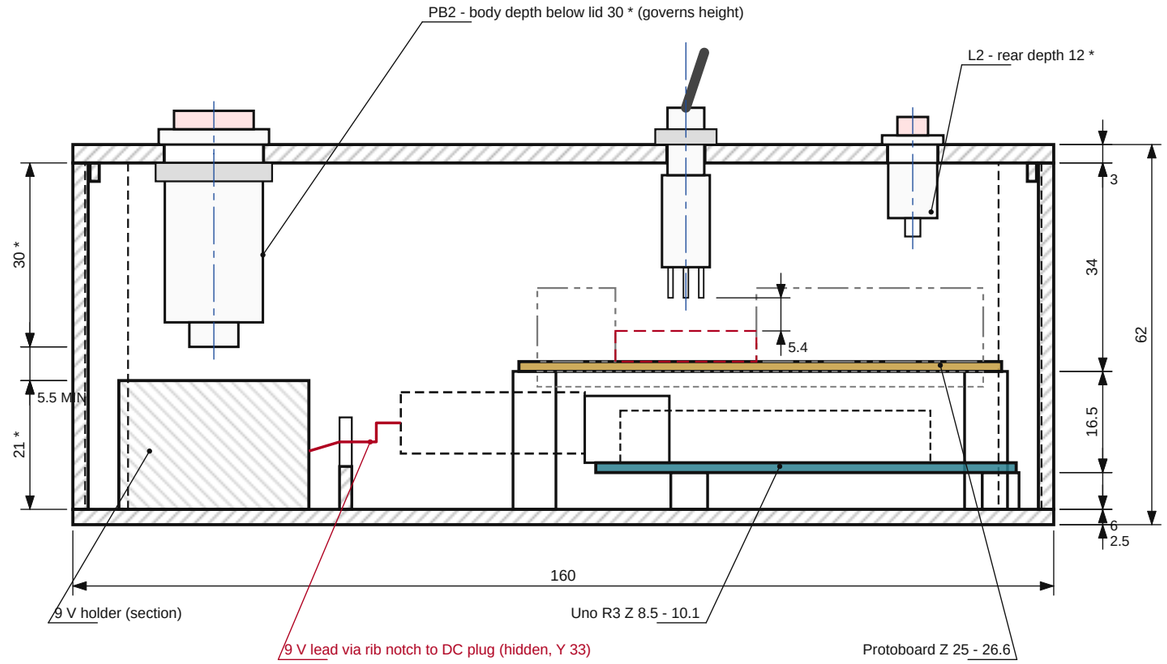
Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: GENERAL ARRANGEMENT - PLAN VIEWS			SHEET: 1 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: C - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

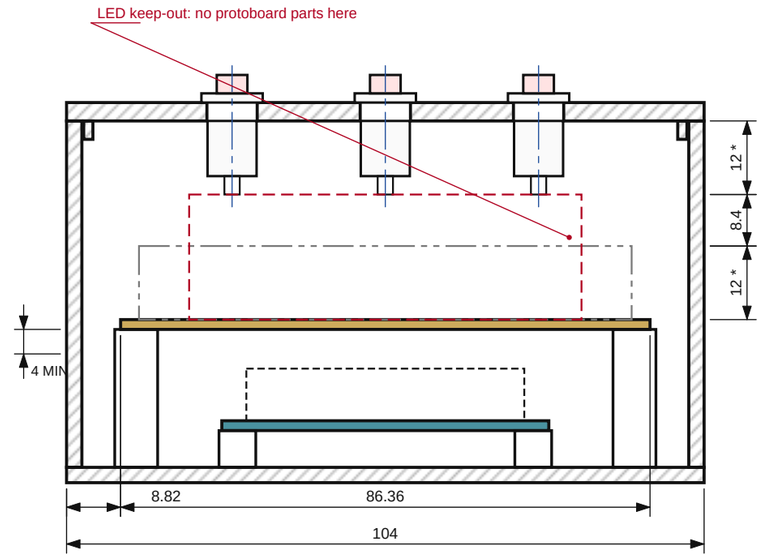
SECTION A-A (stepped, viewed from front)

Stepped section: Y = 52 (PB2, L2), jog to Y = 39.5 through SW1



SECTION B-B (X = 137, viewed from right)

Through LED column L1-L3 and the board stack



ISOMETRIC - EXPLODED LID (front-left view, not to scale)

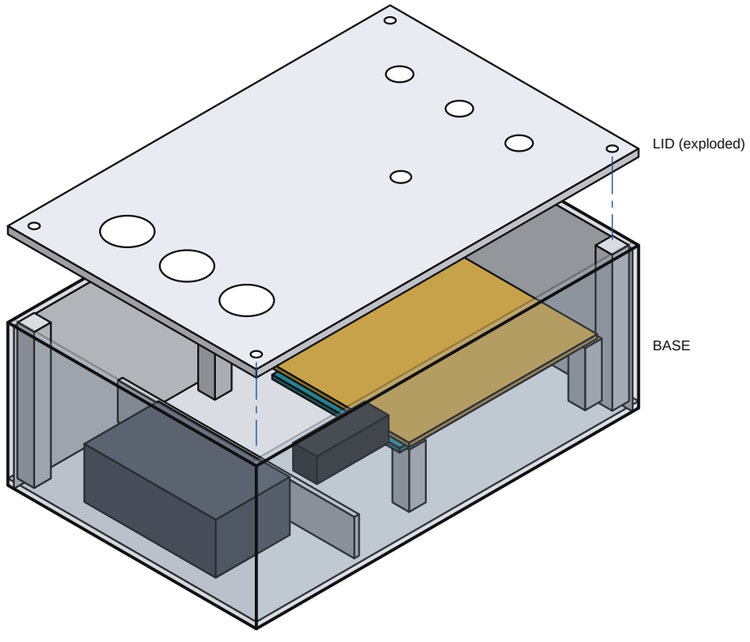


TABLE 2 - VERTICAL STACK-UP (Z from outside bottom)

LAYER	Z RANGE	NOTE
Floor	0 - 2.5	2.5 floor, 4x Ø10 x 1 foot recesses
Uno standoffs	2.5 - 8.5	6.0 high, Ø6
Uno R3 PCB	8.5 - 10.1	1.6 PCB
Uno parts / headers	10.1 - 21.0	DC jack/USB 10.9, headers 8.5
Clearance	21.0 - 25	4 to Uno parts, 3.4 to DC plug
Protoboard	25 - 26.6	on 4 floor pillars, H 22.5
Protoboard parts	26.6 - 38.6	max 12 * (keep-out under LEDs)
LED rear	47 - 59	12 *; 8.4 clear to parts
Battery holder	2.5 - 23.5	21 * high, on floor
Selector rear	37 - 59	22 *; parts under SW1 <= 5
Button rear	29.0 - 59.0	30 *; 5.5 clear to holder
Base top / lid	59.0 - 62.0	3.0 lid, 1.5x3 lip, 0.3 gap

Overall height is set by: holder 21 + 5.5 gap + button 30 + lid 3 + floor 2.5 = 62.

Each 1 mm saved on the measured button depth reduces height by 1 mm.

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 to layout-v6 position.
C 2026-11-09 Uno + 9 V plug per photo. * = assumed.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: SECTIONS A-A / B-B, STACK-UP, ISOMETRIC			SHEET: 2 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: C - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

GENERAL NOTES

1. All dimensions in mm. Third-angle projection. Model origin: external front-left-bottom corner (X right, Y back, Z up).
2. Two printed parts: BASE (tub with floor, walls, rib, bosses, pillars, standoffs) and LID (control panel). Print lid face-down.
3. Walls 2.5, floor 2.5, lid 3.0. External corners R3 (vertical edges) optional; internal fillets R1 at wall/floor joints.
4. Lid fixing: 4x M3 x 10 countersunk screws into M3 x 5.7 heat-set inserts (Ø4.0 pilot, 6 deep) in corner bosses Ø7.
5. Lid lip 1.5 wide x 3.0 deep, 0.3 clearance per side to inner wall; notch lip 8 x 8 around each corner boss.
6. Panel holes are printed at +0.2 over nominal (Ø16.2 / Ø6.2 / Ø8.2). Ream to fit after a test print of a 30 x 30 coupon.
7. SW1 anti-rotation: add a Ø2.5 x 1.5 deep key hole at 8 mm from SW1 centre once the switch's tab washer is confirmed.
8. Protoboard on 4 floor pillars (Ø7 x 22.5) at its corner hole positions, drilled to Ø3.2; M3 self-tapping or insert.
9. Uno R3 on 4 standoffs Ø6 x 6.0 (Ø2.5 pilot for M3 self-tap / nylon M3). USB/DC jack face LEFT (photo).
10. Battery service: remove 4 lid screws, fold lid over to the LEFT on a 150 mm wire service loop; holder is directly accessible. The protoboard / Uno stack is NOT disturbed (spec §6).
11. Wiring: the 9 V lead exits the bay through the 8 x 8 rib notch (Y 48-56), turns along the plug zone floor (tie anchors at Y 25 / 79) and enters the plug, which goes straight into the Uno DC jack at Y 33.13. Deburr all edges.
12. Power per photo: 9 V holder lead ends in a DC barrel plug in the Uno jack. Keep the 38 mm zone left of the jack free for the plug and its cable bend. No USB opening (spec §1). Fit the plug before mounting the protoboard.
13. Protoboard keep-outs: LED column (X 130-144, Y 20-84) parts <= 3; SW1 (X 88.5-111.5, Y 28-51) parts <= 5.
14. Add 4x Ø10 x 1.0 deep recesses on the underside at (12,12) (148,12) (12,92) (148,92) for rubber feet.
15. SW1 position follows the top layout of draft v6 (position only; no v6 dimensions used): right of centre, midway between the L2 and L3 rows. Mount the toggle so the lever throws along Y (back = towards L2, front = towards L3).

TABLE 3 - CLEARANCE CHECKS (computed from model)

CHECK	ACTUAL	MIN	RESULT
Button body -> battery holder (Z)	5.5	5	OK
LED rear -> protoboard parts (Z)	8.4	5	OK
Uno tallest part -> protoboard (Z)	4	2.5	OK
Protoboard edge -> right wall (X)	6	5	OK
DC plug end -> rib (cable bend) (X)	8	8	OK
DC plug top -> protoboard (Z)	3.4	2.5	OK
Protoboard edge -> front/back wall (Y)	6.32	5	OK
Uno edge -> right wall (X)	3.62	3	OK
Selector rear -> protoboard parts (Z)	5.4	5	OK
Button bezel -> bezel (Y)	7	5	OK
Corner boss -> protoboard corner	0.97	0.5	OK

OPEN ITEMS (spec §1, §7)

- External DC input: OPEN, not modelled. No USB opening.
- Confirm button quantity (3) and LED model / quantity (3).
- Confirm 2.8 x 0.5 QC receptacle fit on button blades.
- Verify SW1 terminal arrangement with continuity meter.
- Spec §5 says snap connector; photo shows a DC barrel plug. Drawn per photo.

A right-angle DC plug would cut about 20 mm from the width.

TABLE 4 - ASSUMPTIONS REGISTER (* values: the spec lists these as TBC/approx. - measure with calipers and update the parameters)

ITEM	VALUE USED	BASIS / SOURCE	ACTION
Protoboard outline	78.74 x 86.36 x 1.6	Hole span 73.66 x 81.28 (spec §4) + 2.54 edge margin each side	Measure edge-to-edge
Protoboard part height	12 max above board	Typical radial electrolytic / terminal block	Measure tallest part
Arduino Uno R3	68.58 x 53.34 x 1.6	Arduino reference design; holes (13.97,2.54) (15.24,50.80) (66.04,7.62) (66.04,35.56)	Verify on board
Uno part height	10.9 (USB-B / DC jack)	Reference design; headers 8.5	Verify
9 V battery holder	31 x 55 x 21	Spec §4 'approx. 55 x 31 x 21' - not verified	Measure + wire exit
Battery bay reserve	41 x 65	Spec §4 provisional planning reserve	Re-check after holder
Adafruit 1439 button	Ø16 mount, Ø18 bezel, 30 below panel	16 mm nominal (spec); depth incl. 2.8 mm QC terminal + wire bend ASSUMED	**CRITICAL: sets height**
Miyama MS-500E	Ø6.2 bore, body 7.8 x 13, 22 below panel	Spec §4 listing; depth below panel ASSUMED	Check mfr. drawing
LED indicator	Ø8.2 cut-out, 12 rear depth	Generic 5 mm LED in 8 mm panel holder - model not chosen	Select model
Panel thickness	3.0 lid	Within typical max panel thickness for M6 bushing / 16 mm button	Check both parts
DC plug (9 V lead)	30 long, Ø10, +8 bend	Arrangement from user photo; size estimated - photo not used as a measurement	Measure plug + boot
Clearance allowance	>= 5 (6 at electronics bay)	Spec §6 preliminary allowance; 6 used so corner bosses clear the board	-

HOW TO UPDATE AFTER MEASURING

a) Edit the parameter block at the top of cad/power_grid_box_enclosure.scad (and tools/gen_drawing.py for this drawing set).
b) Overall size follows automatically: width = wall + battery bay + rib + DC plug zone (plug + bend) + Uno + 3 + wall;
depth = walls + 2 x 6 + protoboard depth; height = floor + holder + 5 + button depth + lid (re-check Table 3).
c) Re-run tools/gen_drawing.py - the clearance table (Table 3) and the asserts fail if any check drops below its minimum.
d) Before the full print: print the lid only (approx. 1.5 h) and trial-fit PB1-3, SW1 and L1-3.

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 to layout-v6 position.
C 2026-11-09 Uno + 9 V plug per photo. * = assumed.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE	DRAWING No.: PGB-ENC-001
TITLE: NOTES, ASSUMPTIONS REGISTER, VERIFICATION	SHEET: 3 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: C - PRELIM.
DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated
STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	